

BCSE325L	INTRODUCTION TO BITCOIN		L	T	P	C
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Pre-requisite	NIL	Syllabus version				
		1.0				
Course Objectives						
1. To Identify the process of Cryptocurrency.						
2. To understand the functionality of Bitcoin.						
3. To explore the recent developments on Bitcoin.						
Course Outcomes						
After completion of this course, the student shall be able to:						
1. Understand the fundamentals of Cryptography.						
2. Gain knowledge about various operations associated with Cryptocurrency.						
3. Develop the methods for verification and validation of Bitcoin transactions.						
4. Apply the principles, practices and policies associated with Bitcoin business.						
Module:1	Fundamentals of Cryptography				5 hours	
Cryptographic Hash Functions - Hash Pointers and Data Structures - Digital Signatures - Public Keys as Identities - A Simple Cryptocurrency.						
Module:2	Features of Bitcoin				6 hours	
Bitcoin Transactions - Bitcoin Scripts - Applications of Bitcoin Scripts - Bitcoin Blocks - Bitcoin Network and Limitations.						
Module:3	Bitcoin Techniques				7 hours	
Techniques to Store and Use Bitcoins - Hot and Cold Storage - Splitting and Sharing Keys - Online Wallets and Exchanges - Payment Services - Transaction Fees - Bitcoin Trading.						
Module:4	Bitcoin Mining				8 hours	
Task of Bitcoin Miners - Mining Hardware - Energy Consumption and Ecology - Mining Pools - Mining Incentives - Merkle Tree - hardness of mining - transaction verifiability.						
Module:5	Bitcoin and Anonymity				5 hours	
Anonymity – Re-identification of Bitcoin - Mixing and Decentralisation of Bitcoin - Zero coin and Zero cash.						
Module:6	Mining Strategies				5 hours	
Essential Puzzle Requirements – Application Specific Integrated Circuit Resistant(ASIC) Puzzles - Proof of Volunteer computing - Non externalization of Puzzles - Proof of Stake Virtual Mining.						
Module:7	Bitcoin as a Platform				7 hours	
Bitcoin as an Append-Only Log - Bitcoin as Smart Property - Secure Multi-Party Lotteries in Bitcoin - Bitcoin as Randomness Source - Prediction Markets and Real-World Data Feeds.						
Module:8	Contemporary Issues				2 hours	
	Total Lecture hours:				45 hours	
Text Book						
1.	Goldfeder, S., Bonneau, J., Miller, A., Felten, E., Narayanan, A. Bitcoin and Cryptocurrency Technologies, 2016, 1st edition, Princeton University Press, New Jersey.					
Reference Books						
1.	Antonopoulos, A. M. Mastering Bitcoin: unlocking digital cryptocurrencies, 2017, 2 nd edition, O'Reilly Media, Inc, United States.					
2.	Lewis, Antony, The Basics Of Bitcoins and Blockchains: An Introduction To Cryptocurrencies and The Technology That Powers Them., 2018, 1 st edition, Mango Media Inc., United States.					
Mode of Evaluation: CAT / Assignment / Quiz / FAT						
Recommended by Board of Studies			04-03-2022			
Approved by Academic Council			No. 65	Date	17-03-2022	